DSAA 1001 Project: Chicago Food Inspection Data Analysis

Exploratory Data Analysis + Binary Classification

1. Project Overview

Imagine you are helping the City of Chicago plan food safety inspections.

Before an inspector visits a food establishment, some information is already known: what kind of place it is, why it is being inspected, and what happened in earlier inspections. What we do not know yet is the result of the upcoming inspection.

Can we use what is known before the visit to predict whether this inspection will Fail?

You will first explore the historical data to understand what it contains, identify data quality problems, and find patterns that may be useful for prediction. Then you will turn those findings into features and build a classification model.

Core task: Using only information available before an inspection begins, predict whether that inspection will result in Fail.

You should form a team of up to 3 students by 1 Oct 2026.

2. What Is the Data?

The project uses the City of Chicago Food Inspections dataset (dataset ID: 4ijn-s7e5). Each row represents one inspection. The same establishment may appear more than once if it has been inspected multiple times.

Official dataset page: City of Chicago - Food Inspections

Historical data period: 2016-01-01 to 2026-10-30. You may use records from this period that were publicly available by 2026-10-30 23:59. The teaching team will provide an initial snapshot, and you may supplement it with records downloaded from the portal before the cutoff. Save the data version used in your analysis and record the source, download date, row count, and missing-value rates in README.md.

Main fields

Field(s)	What it describes
Inspection ID	Unique inspection identifier; used to match predictions to test records.
DBA Name / AKA Name / License #	Business identity information.
Facility Type / Risk	Type of establishment and recorded risk category.
Address / City / State / Zip / Latitude / Longitude / Location	Location information.
Inspection Date / Inspection Type	When and why the inspection took place.
Results	Inspection result; used to define the prediction target.
Violations	Violations found during the inspection, with descriptions/comments.

3. Fields, Features, and Target

These terms are related but not the same:

- **Raw field:** A column in the original dataset.
- **Feature:** Information used as model input. A feature may come directly from a field or be constructed from available information.
- **Target:** The value the model is trying to predict. In this project, the target is derived from Results.

All features must follow the main rule: they can only use information available before the inspection begins. Information produced by the current inspection, such as its final result or violations, cannot be used to predict that same inspection.

4. Prediction Task and Labels

Results value	Label	Use in this project
Fail	1	Positive class
Pass	0	Negative class
Pass w/ Conditions	0	Negative class
Out of Business / No Entry / Not Ready / Business Not Located	-	Not included in labelled training or evaluation data

Records without a Pass/Fail judgement are not labelled examples for training or evaluation. If such a record occurred before the inspection being predicted, it may still be used to describe that establishment's history.

5. Hidden Test Set

The hidden test period is 2026-10-31 to 2026-11-27. The teaching team will provide the test records with the fields available before inspection, while the Results labels will remain hidden. Do not query the portal for results from this period, and do not use hidden test outcomes for training, feature construction, or tuning.

6. Evaluation Metric

The main metric is Macro-F1. F1 is calculated separately for class 0 and class 1 and then averaged, so both classes receive equal weight in the final score.

Leaderboard calculation: f1_score(y_true, y_pred, labels=[0, 1], average="macro", zero_division=0)

7. Prediction File

Submit predictions.csv with one row for each hidden test record:

```
Inspection ID,pred_fail
2734512,0
2734518,1
```

pred_fail must be an integer label: 1 = predicted Fail, 0 = predicted non-Fail. If your model outputs probabilities, you may use the default threshold or adjust it on the validation set. Record the chosen threshold in the notebook and lock it before final evaluation.

8. Assessment 1 - EDA Notebook

Due: 30 Oct 2026 (Fri), 23:59

Submit one .ipynb notebook that runs with Run All, uses relative paths, works offline, and explains the analysis in Markdown cells.

- **Exploratory analysis and figures:** Show findings that are useful for later modelling. Around 6-8 figures is a suggested range; adjust as needed. Each figure should clearly show what is being compared, what you found, and why it matters.
- **Data quality analysis:** Identify problems in the data and support them with specific evidence. You may summarise them using columns such as: problem, evidence, possible reason, and effect on modelling.
- **Modelling plan:** In about one page, describe your baseline model and evaluation plan, including how time will be split and how repeated inspections of the same establishment will be considered.

9. Assessment 2 - Final Notebook and Predictions

Due: 4 Dec 2026 (Fri), 23:59

Submit the final .ipynb notebook and predictions.csv, and complete one presentation option below. The final notebook must reproduce the analysis and generate predictions.csv from the saved raw data and test input using one command or Run All. The teaching team will run the repository offline from a fresh clone. Set random seeds where appropriate.

- **Phase 1:** Revised EDA and data-quality analysis.
- **Phase 2:** Model training and evaluation.
- **Final results and conclusions:** Summarise performance, findings, and limitations.
- **Analysis brief:** About one page for the City Health Commissioner, focusing on main findings, implications for inspection planning, and limitations.

10. Modelling and Evaluation Requirements

- Use earlier records for training and later records for validation and testing. Cross-validation, model selection, and threshold adjustment must respect time order. Your own held-out test set is for final evaluation only.
- When designing the split, consider repeated inspections of the same establishment. Explain whether your evaluation mainly measures later inspections of known establishments, unseen establishments, or a mixture of both. You may add grouped evaluation by establishment. TimeSeriesSplit is one possible option, but it does not automatically isolate establishment groups.
- You may improve the provided baseline through features, processing, or model settings, or use other classifiers such as Logistic Regression, Random Forest, XGBoost, or LightGBM. Complex models are not required.
- On the held-out test set, report Macro-F1, precision, recall, and F1 for each class, plus a confusion matrix. Discuss what the two types of error mean in practice. You may also include other metrics if they help interpret the model.
- Discuss limitations revealed by the data, evaluation design, or experiments and explain how they affect practical interpretation.

11. Presentation Options

Choose one presentation option:
• Option 1 - In person: Have the TAs review your project during the last lab session.
• Option 2 - Video: Submit a video showing your execution results and clearly explaining your code.
For either option, each group member must state their name and present the part they were responsible for. For the video option, submit an .mp4 or an accessible link with clear audio and readable on-screen content.

Late submissions will not be accepted.

12. Marking Scheme

Component	Marks
Phase 1: EDA and Data Quality	7
Phase 2: Modelling, Evaluation, and Communication	10
Code Integrity	3
Bonus	Up to 5 (overall project mark capped at 20)

Phase 1 - 7 marks

Assess data quality and structure and explain how findings affect modelling. Useful starting points include missing values, data types, duplicates, category consistency, changes over time, differences across areas/facility types/inspection types, repeated inspections, and whether fields are available at prediction time. Figures are judged on clarity, appropriate presentation, and whether they support the written interpretation.

Phase 2 - 10 marks

Build and evaluate the classifier, explain modelling choices and improvements over the baseline, report the required evaluation results, interpret errors, and discuss limitations and practical implications.

Code Integrity - 3 marks

- Reproducibility: the notebook must run offline from a fresh clone, without manual steps, and generate predictions.csv. If it does not run, this component receives 0.
- Data use: all project inputs must be saved in the repository with source and download date. External data such as census, weather, licence, or boundary data may be used if they satisfy the time boundary and are available at prediction time.
- AI use: AI coding assistants are allowed. Record the tools used and what they were used for in AI-USAGE.md. Every group member must be able to explain their submitted work, including AI-assisted work.

13. Bonus (Optional)

You may attempt one or both bonus components. Their combined contribution is capped at 5 marks, and the final project mark is capped at 20.

- **Depth of analysis:** Up to 5 marks. Each well-supported deeper finding is worth 1 mark (up to 4), with +1 for one valid and unexpected finding.
- **Leaderboard:** Based on hidden-test Macro-F1: ranks 1-2 = 5; 3-5 = 4; 6-9 = 3; 10-14 = 2; 15-20 = 1. Invalid submissions or submissions not exceeding the published baseline receive 0. Using post-cutoff results, hidden labels, or information unavailable at prediction time also results in 0 leaderboard bonus.

14. Questions

Post questions about the data, requirements, or marking on the course forum so that all groups can receive the same clarification.